

ROCKET AND MISSILES

PROFESSIONAL ELECTIVE – V

VIII Semester										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE54		PCC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I	ROCKET PROPELLANT SYSTEMS									
SOLID PROPELLANT ROCKET SYSTEMS: Ignition system in rockets, Types of igniters, Igniter design considerations, Combustion system of solid rockets.										
LIQUID PROPELLANT ROCKET SYSTEMS: Design consideration of liquid rocket combustion chamber, injector, propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems, Propellant slosh, Propellant hammer, Geysering effect in cryogenic rocket engines										
UNIT-II	AERODYNAMICS OF ROCKETS AND MISSILES									
Classification of missiles, general aerodynamic design considerations, Airframe components of rockets and missiles and aerodynamic characteristics, Forces acting on a missile while passing through atmosphere, Method of describing aerodynamic forces and moments, Drag estimation, Body upwash and downwash in missiles, free flight dispersion and structural design considerations										
UNIT-III	TWO-DIMENSIONAL ROCKET MOTION IN VACUUM & MULTI-STAGE ROCKET									
TWO-DIMENSIONAL ROCKET MOTION IN VACUUM: Equations of motion, Rocket Motion in free space (Tsiolkovsky's equation, Rocket Parameters, Burnout range), Rocket Motion in a homogeneous gravitational field (Vertical flight).										
MULTI-STAGE ROCKET: Nomenclature of the multi-stage rocket, Ideal Velocity of the multi-stage rocket, Vertical ascent in a homogeneous gravitational field and in vacuum (Burnout velocity- Culmination altitude- Vertical ascent of a two-stage rocket).										
UNIT-IV	ATTITUDE CONTROL OF ROCKETS AND MISSILES									
ATTITUDE CONTROL OF ROCKETS AND MISSILES: Rocket thrust vector control, Methods of thrust vector control, Thrust magnitude control, Thrust Termination.										
SEPARATION SYSTEMS FOR ROCKETS AND MISSILES: Stage separation dynamics, Separation and aerodynamics launching problems.										
UNIT-V	MATERIALS FOR ROCKETS AND MISSILES									
Criteria for Selection of materials for rockets and missiles, Choice of materials at cryogenic temperatures, extremely high temperatures. Requirement of materials for thermal protection and pressure vessels.										
Text Books:										
1. Martin J. L. Turner (2008), Rocket and Spacecraft Propulsion principles, practice and new developments, 3rd edition, Springer, USA.										
2. Sutton G.P. (2010), Rocket Propulsion Elements, John Wiley / BSP Books, USA.										
3. Cornelisse J. W. (1980), Rocket Propulsion and Space Dynamics, Pitman Publishing, London.										
Reference Books:										
1. S. S. Chin (1982), Missile Configuration Design, McGraw- Hill, New Delhi.										
2. Bong Wie (2008), Space Vehicle Dynamics and Control, AIAA Educational Series, USA.										
3. Earl R Parker (1998), Materials for Missiles and Spacecraft, McGraw Hill, New Delhi										

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Interpret new technology for staging of rockets and missiles
2. Develop methods for altitude control.
3. Determine 2-D motion rocket equation.
4. Analyse the missile guidance system.
5. Identify material for nozzle of rockets and missiles.